

Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

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Abstract—Automatic crack extraction is essential for both infrastructure inspection and geosciences, but annotated crack data remain scarce and domain shifts are frequent. We address the problem of extracting a crack network from one or several scalar modalities, without training.

To do so, we generalize the classical FRANGI vesselness filter from a pixel-wise operator to a pairwise spatial graph. While the standard filter is limited to independent pixel-wise geometry, our formulation enriches this description by explicitly encoding the orientation alignment between neighboring pixels, alongside elongation and contrast. The final crack network is then extracted through a graph pipeline combining minimum spanning tree reduction and similarity-weighted tree centrality.

While supervised models perform best on clean, in-distribution data, our geometry-driven framework degrades more slowly than a state-of-the-art deep-learning method under the tested synthetic perturbations.

Index Terms—Crack detection, multi-modal fusion, FRANGI filter, graph, minimum spanning tree, tree centrality

I. INTRODUCTION

Reliable crack mapping is needed for civil infrastructure inspection and for monitoring geological hazards such as landslides, faults and subsidence [1]. In practice, many crack maps are still produced or corrected manually, which is slow, subjective and difficult to scale. Automatic extraction is difficult because cracks are thin, discontinuous, and are often low-contrast structures whose appearance varies with the sensor, material, illumination, surface roughness and acquisition geometry.

Deep learning methods have become the dominant approach for crack segmentation. They perform well when training and test images follow the same distribution, but they require annotated data and may fail under sensor or domain shifts [2]–[4]. This limitation is especially relevant for multi-modal crack analysis, where public benchmarks with clean, well-aligned visible, depth or thermal modalities remain rare.

This paper takes a complementary direction: Our framework provides an interpretable, training-free extraction pipeline. It addresses multi-modal scenarios for which annotations are not available. In this setting, creating ground truth is very expensive. Our method can directly assist domain experts (*e.g.*, geophysicists) and accelerate dataset annotation.

Our main contributions are as follows: (1) We generalize the FRANGI filter [5] from a pixel-wise vesselness response to a pairwise graph similarity using second-order shape, contrast and alignment terms, together with normalized multi-modal

Hessian fusion; (2) We evaluate our complete algorithm on the FIND benchmark [6] and on two geological image sets, including comparisons with supervised models on clean and noisy data. The contribution lies in coupling these established components through the pairwise FRANGI graph.

II. RELATED WORK

A. Classical Curvilinear Filters

Multi-scale Hessian filters are a standard tool for enhancing curvilinear structures. The classical FRANGI vesselness filter [5] uses the eigensystem of a smoothed Hessian matrix to assess whether a pixel belongs to a tubular structure, following related work such as SATO's line filter [7]. These methods are fast and interpretable, but they are local: They evaluate pixels independently and can fragment under noise or textured backgrounds. Our method keeps the Hessian geometry but moves the decision from isolated pixels to edges of a spatial graph. Our preliminary study [8] considered two single-modal geological images. Here we add normalized multi-modal fusion, evaluate clean and noisy FIND data, and include a second geological setting.

B. Deep Crack Segmentation

Deep learning architectures such as U-Net [9], DeepCrack [10], CrackFormer [11] and recent diffusion models such as CrackSegDiff [12] have achieved strong crack segmentation results. Their main practical weakness is their dependence on annotations. A recent survey by ZHANG *et al.* [4] also emphasizes poor cross-domain robustness when sensors, materials or acquisition conditions change. This motivates training-free alternatives and hybrid strategies that can operate when annotations are not available.

C. Multi-Modal Fusion

In machine learning, multi-modal fusion is usually implemented by channel concatenation, intermediate feature fusion or late prediction fusion. These choices depend on the network architecture and on the modalities available at training time. Here, fusion is performed at the operator level: Each modality contributes a normalized Hessian, and the fused Hessian is processed by the same graph pipeline regardless of the number of modalities.

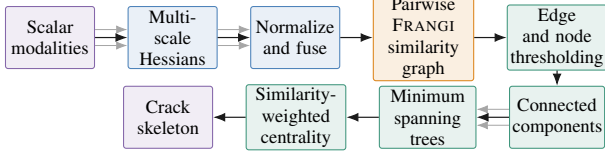


Fig. 1: Method overview. Colors follow Secs. III-A–D (purple/blue/orange/green); Sec. III-E summarizes their roles. Parallel black/gray arrows denote multiple modalities and retained components.

III. METHODOLOGY

A. Problem Formulation and Skeleton Output

Let $\Omega \subset \mathbb{Z}^2$ be the image grid and let

$$\mathcal{I} = \{I^{(m)} : \Omega \rightarrow \mathbb{R}\}_{m \in \mathcal{M}} \quad (1)$$

be one or several scalar modalities, such as visible intensity, depth/range or thermal infrared images. The output is a set of pixels $\hat{\Gamma} \subset \Omega$ representing the crack skeleton. We use “skeleton” in the standard image-analysis sense: A one-pixel-wide centerline representation that preserves the main connectivity and branches of a thin object; thick masks used for evaluation are skeletonized with the implementation reported in [13], [14].

The method assumes that the crack of interest is visible as a dark curvilinear ridge. If a sensor encodes cracks as bright ridges, the modality can be inverted before processing. Figure 1 summarizes the processing chain.

B. Multi-Scale Hessian Features

For a modality $I^{(m)}$ and a Gaussian scale σ , the smoothed Hessian at $x \in \Omega$ is

$$\mathcal{H}_\sigma^{(m)}(x) = \begin{pmatrix} \partial_{xx}(G_\sigma * I^{(m)})(x) & \partial_{xy}(G_\sigma * I^{(m)})(x) \\ \partial_{xy}(G_\sigma * I^{(m)})(x) & \partial_{yy}(G_\sigma * I^{(m)})(x) \end{pmatrix}. \quad (2)$$

Equivalently, the derivatives are computed by convolution with derivatives of the Gaussian kernel [5], [15],

$$\partial_{ab}(G_\sigma * I^{(m)}) = (\partial_{ab}G_\sigma) * I^{(m)}, \quad a, b \in \{x, y\}. \quad (3)$$

Since the Hessian matrix in (2) is real and symmetric, we compute its eigenvalues $\lambda_1^\sigma(x), \lambda_2^\sigma(x)$, ordered as $|\lambda_1^\sigma(x)| \leq |\lambda_2^\sigma(x)|$. The eigenvector associated with $\lambda_1^\sigma(x)$ gives the local longitudinal direction of the ridge. For an elongated ridge, transverse curvature dominates longitudinal curvature, so $R_B^\sigma(x) = |\lambda_1^\sigma(x)|/|\lambda_2^\sigma(x)|$ is near zero; for a blob-like structure, the two magnitudes are comparable and $R_B^\sigma(x)$ approaches one. Classical FRANGI vesselness combines this geometry with curvature magnitude and sign at each pixel. Our graph retains these local cues while requiring compatible evidence and longitudinal directions at neighboring pixels.

For multi-modal inputs, each Hessian is normalized by the largest spectral norm [16] in the image,

$$\hat{\mathcal{H}}_\sigma^{(m)}(x) = \frac{\mathcal{H}_\sigma^{(m)}(x)}{\max_{y \in \Omega} \left\| \mathcal{H}_\sigma^{(m)}(y) \right\|_2}, \quad (4)$$

and the fused Hessian is

$$\mathcal{H}_\sigma^{\text{fused}}(x) = \sum_{m \in \mathcal{M}} w_m \hat{\mathcal{H}}_\sigma^{(m)}(x), \quad w_m \geq 0, \quad \sum_m w_m = 1. \quad (5)$$

All eigenvalues and orientations used below are computed from (5).

We assume that the crack geometry is shared across modalities, whereas sensor-specific artifacts are largely uncorrelated. Under this assumption, averaging the normalized Hessians reinforces consistent structures and reduces modality-specific noise.

C. Generalized FRANGI Similarity Graph

We build a sparse spatial graph $G = (V, E)$ on candidate pixels $V \subseteq \Omega$. We denote the distance between two candidate pixels by

$$\rho_{ij} = \|x_i - x_j\|_2. \quad (6)$$

Pixels $x_i, x_j \in V$ are connected when

$$0 < \rho_{ij} \leq R. \quad (7)$$

The radius R allows short interruptions caused by noise or local loss of contrast to be bridged, while ρ_{ij} discourages long accidental jumps.

For a scale $\sigma \in \Sigma$, an edge (i, j) is assigned a base similarity $S_{ij}^\sigma \in [0, 1]$ as a product of three soft constraints.

Curvature evidence. For dark ridges, the useful transverse curvature is $C_\sigma(x) = \max\{\lambda_2^\sigma(x), 0\}$. The intensity term is

$$S_{\text{int}}^\sigma(i, j) = 1 - \exp \left[-\frac{1}{2} \left(\frac{\sqrt{C_\sigma(x_i)C_\sigma(x_j)}}{s_i} \right)^2 \right]. \quad (8)$$

It requires strong positive transverse curvature at both endpoints; the geometric mean makes the edge weak when either endpoint lacks this evidence. This term carries the dominant information of the classical FRANGI response.

Shape refinement. On the structures detected by S_{int}^σ , both endpoints should also have a small FRANGI ratio. Blob-like responses are therefore down-weighted by

$$S_{\text{shape}}^\sigma(i, j) = \exp \left[-\frac{1}{2} \left(\frac{R_B^\sigma(x_i) + R_B^\sigma(x_j)}{s_s} \right)^2 \right]. \quad (9)$$

Alignment refinement. If $\theta^\sigma(x)$ is the angle of the longitudinal eigenvector, locally incompatible directions on that same curvature support are down-weighted by

$$S_{\text{align}}^\sigma(i, j) = \exp \left[-\frac{1}{2} \left(\frac{\sin(\theta^\sigma(x_i) - \theta^\sigma(x_j))}{s_a} \right)^2 \right]. \quad (10)$$

The sine accounts for the fact that crack directions are unoriented: θ and $\theta + \pi$ describe the same local axis.

Thus S_{shape}^σ and S_{align}^σ refine, rather than replace, the curvature structures identified by S_{int}^σ . We first take the strongest base similarity across scales,

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^\sigma(i, j) S_{\text{shape}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j), \quad (11)$$

which compares the local appearance and geometry without yet penalizing separation. We then define the graph dissimilarity as the $[0, 1]$ -clipped value of $\rho_{ij}(1 - S_{ij}^{(0)})$, and its complementary similarity:

$$\begin{aligned} d_{ij} &= \text{clip}_{[0,1]}(\rho_{ij}(1 - S_{ij}^{(0)})) \\ &= \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}. \end{aligned} \quad (12)$$

For unit-distance neighbors ($\rho_{ij} = 1$), dissimilarity is simply the complement of the base similarity. Across a short gap, $\rho_{ij} > 1$ raises the cost, discouraging an accidental long connection while still allowing a strongly supported interruption to be bridged; clipping keeps the result bounded. In the subsequent pipeline, S_{ij} is used for graph thresholding and centrality, whereas d_{ij} supplies the costs for the subsequent minimum spanning trees.

For FIND and the Palais des Papes (*cf.* Sec. IV-A), we use $s_s = 2$, $s_i = 1/4$, $s_a = 1/8$, $\Sigma = \{1, 3, 5, 7\}$ and $R = 3$. For the Vaches Noires images (*cf.* Sec. IV-A), the single-modal implementation uses $s_s = 1/2$, $s_i = 1/4$, $s_a = 1/8$, $\Sigma = \{1, 3, 5, 7, 9\}$ to cover wider visible fractures, with the same radius $R = 3$ as in the preliminary study.

D. Graph Extraction

The extraction pipeline consists of four steps. First, edge thresholding keeps the strongest fraction τ_E of edges, ranked by S_{ij} . On this retained edge set, each node is scored by its maximum incident similarity; node thresholding then keeps the strongest fraction τ_V of nodes. To reduce tuning, all experiments tie the two retention factors, $\tau_E = \tau_V = \tau$.

Second, connected components are obtained from the remaining graph. FIND provides the prior that only one crack structure is of interest, so we retain the component with the largest number of pixels. Without this prior, we retain every component whose pixel count exceeds a prescribed fraction $\eta|\Omega|$ of the image pixels; η is chosen from the minimum size of the structures one wishes to extract. The next two steps are applied separately to every retained component.

Third, a minimum spanning tree $\mathcal{T}$ [17] is computed for each retained component using the dissimilarity d_{ij} . Each tree preserves the “cheapest” coherent connections and turns the locally thick response into a structure suitable for centerline extraction.

Fourth, we compute a similarity-weighted tree centrality [18], unlike general-graph betweenness [19]. It takes $O(|V|)$: Breadth-first traversal records parents level by level; its reverse order accumulates subtree masses. Each vertex receives the weight

$$w_i = \max_{j:(i,j) \in E} S_{ij}. \quad (13)$$

After rooting the tree, let $\mathcal{T}_i$ be the subtree below x_i , $m_i = \sum_{x_k \in \mathcal{T}_i} w_k$, and $M = \sum_k w_k$. We use

$$C_{\mathcal{T}}(x_i) = m_i(M - m_i). \quad (14)$$

The highest-scoring vertex becomes the tree root, and the subtree masses are recomputed from it. The non-root scores

follow (14) and are normalized, while the selected root is assigned score one. A branch is stopped when a child falls below the relative or absolute cutoff. Pixels on the main axis therefore remain, while short peripheral branches are removed.

Figure 2 shows the intermediate maps for a FIND example. The positive-curvature map is close to a classical pixel-wise FRANGI response; the graph similarity is cleaner because it also enforces pairwise orientation compatibility.

E. Role of the Components

The construction starts with S_{int} , which imposes strong positive transverse curvature λ_2 at both endpoints and contains the dominant FRANGI evidence. The other pairwise terms act on these detected structures as refinements: They discourage shapes and connections that do not match the crack prior. Table I summarizes this progression.

TABLE I: The FRANGI evidence and its graph refinements.

Component	Role in the extraction chain
S_{int}	Detects strong positive transverse curvature at both endpoints; provides the dominant classical FRANGI evidence.
S_{shape}	On that support, penalizes blob-like responses and favors elongated structures.
S_{align}	On that support, penalizes incompatible directions and accidental bridges.
ρ_{ij}	In d_{ij} , discourages long links between otherwise similar pixels.
Weighted centrality	Selects pixels lying on the main centerline of the retained graph.

The sensitivity parameters were selected empirically, starting from our preliminary study [8]. We retained s_i and s_a and increased s_s from $1/2$ to 2 for FIND and the Palais des Papes, allowing more variation in the FRANGI ratio, which is less stable than the curvature and orientation terms.

The scale set, radius and tied retention factor are fixed per dataset. We set Σ from the expected crack-width range and R from the largest gap that should be bridged; $\tau = \tau_E = \tau_V$ controls the sparsity–recall trade-off. We use $\tau = 0.25$ for FIND and the Palais des Papes, and $\tau = 0.30$ for Vaches Noires, following the preliminary study [8].

IV. EXPERIMENTS

A. Datasets and Protocol

The quantitative evaluation uses FIND [6], a public benchmark with aligned intensity (I) and range (R) images for road-crack detection (and a third modality, “filtered” F , which is a clean post-processing of the range modality). We compare against CrackSegDiff [12], a supervised diffusion model for multi-modal crack segmentation. For both compared methods, the same evaluation protocol is used: Dense masks are skeletonized before comparison and overlap metrics are computed with a fixed spatial tolerance.

We also report experiments on two geological image sets provided by Cerema. The first contains two annotated 512×512 visible UAV patches from the Vaches Noires landslide at Villers-sur-Mer (France). We call these evaluation patches

Test1 and Test2; they are separate from the 3760×2058 image used to adapt the U-Net baseline. The second set is a photogrammetric intensity-range pair acquired on the retaining wall of the Palais des Papes in Avignon (France).

B. Metrics

For a ground truth set A and a predicted set B , we use the JACCARD index [20], also known as Intersection over Union [21],

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad (15)$$

and the TVERSKY index [22],

$$T(A, B) = \frac{|A \cap B|}{|A \cap B| + \alpha |A \setminus B| + \beta |B \setminus A|}, \quad (16)$$

with $\alpha = 1$ and $\beta = 0.5$, so missed cracks are penalized more than false alarms. Since skeletons are one-pixel-wide, we dilate both skeletons by 3 pixels before computing these overlap scores. This tolerance is empirical and only compensates for small localization shifts at the 256×256 FIND resolution. We also report a discrete WASSERSTEIN distance [23], [24], which measures spatial displacement between predicted and reference skeleton pixels.

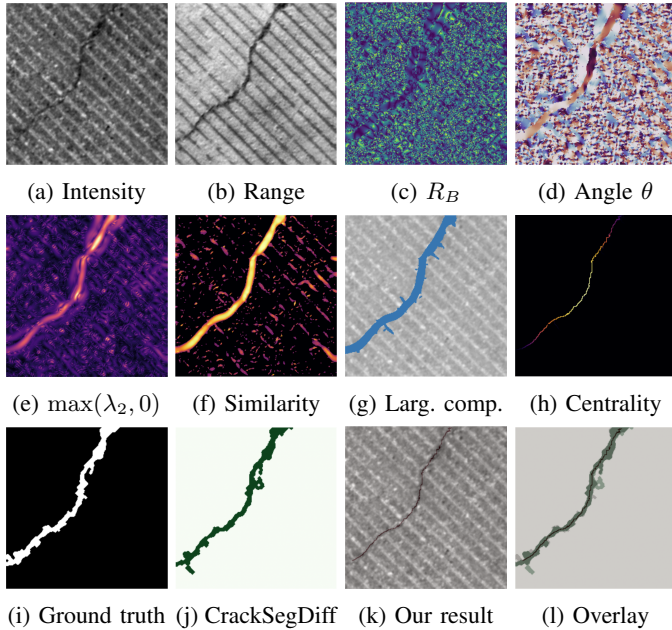


Fig. 2: FIND example. **Top:** Intensity, Range, FRANGI ratio and orientation. **Middle:** Positive curvature, similarity, largest component and weighted centrality. **Bottom:** Ground truth, CrackSegDiff, ours (63% JACCARD, 71% TVERSKY, 8 px WASSERSTEIN) and overlay. At each $x \in V$, $|\lambda_2|$ selects the scale σ at which R_B , θ and $\max(\lambda_2, 0)$ are reported; pairwise similarity is the maximum over incident edges, each already maximized over σ .

C. FIND: Clean and Noisy Data

Table II compares the proposed method with CrackSegDiff on clean FIND images using intensity and range. As expected, CrackSegDiff achieves the highest overlap on the clean, in-distribution data it was trained on. However, our geometric formulation still recovers the main topology of the fracture networks without requiring any training data, demonstrating its viability as a training-free baseline.

TABLE II: Performance on clean FIND data

Method	JACCARD	TVERSKY	WASSERSTEIN
Ours ($I + R$)	60%	68%	18 px
CrackSegDiff	87%	90%	5 px

The FIND dataset [6] also contains a filtered range modality. Table III reports a modality contribution analysis for the proposed method. The filtered modality alone is strong, which shows the importance of clean, possibly pre-processed inputs for our method. Fusing intensity, range and filtered range gives the best result and supports the operator-level fusion strategy.

To evaluate the framework's robustness to synthetic sensor noise, we artificially degrade the input modalities. Specifically, we perturb the intensity modality with multiplicative speckle noise to simulate surface reflectance variations, and corrupt the range modality with additive Gaussian noise to model instrumental measurement errors:

$$I_{\text{noisy}}(x) = I(x) \cdot (1 + N(x)), \quad (17)$$

$$R_{\text{noisy}}(x) = R(x) + N'(x), \quad (18)$$

where $I(x)$ and $R(x)$ denote the original intensity and range signals at pixel x , respectively. $N(x), N'(x) \sim \mathcal{N}(0, s^2)$ are independent Gaussian random variables where s^2 measures the variance of the noise. This protocol evaluates the robustness of our operator-level fusion to controlled sensor perturbations.

Figure 3 reports the degradation curves, and Figure 4 shows an example at a strong noise level. CrackSegDiff initially performs better, but its score decreases rapidly as noise increases. The proposed method starts lower but degrades more smoothly because the graph uses multi-scale curvature and orientation compatibility instead of learned appearance statistics alone.

D. Geological Domain Shift

The Vaches Noires data differ markedly from the road-crack FIND images and contain only a visible modality. We use

TABLE III: Modality contribution analysis for the proposed method on FIND. I : Intensity, R : Range, F : Filtered range.

Weights	JACCARD $\uparrow$	TVERSKY $\uparrow$	WASSERSTEIN $\downarrow$
$I = 1$	41%	48%	43 px
$R = 1$	54%	62%	20 px
$F = 1$	60%	68%	12 px
$I = \frac{1}{2}, R = \frac{1}{2}$	60%	68%	18 px
$I = \frac{1}{3}, R = \frac{1}{3}, F = \frac{1}{3}$	63%	71%	11 px

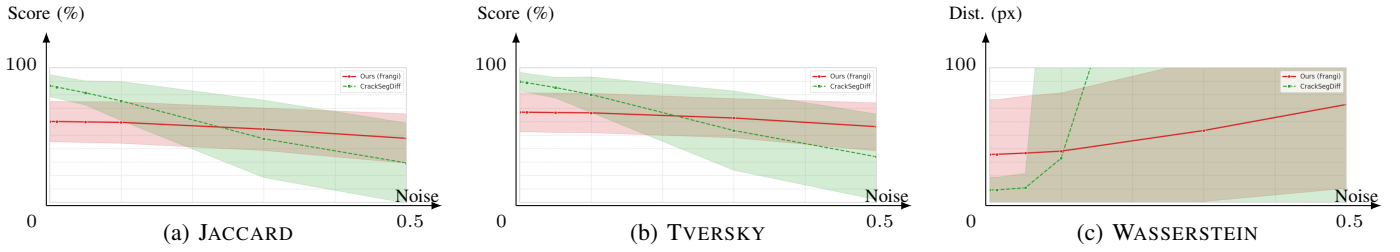


Fig. 3: Performance degradation on noisy FIND. Red: Proposed method; Green: CrackSegDiff. Shaded regions show standard deviation. The horizontal axis is the noise level s (standard deviation of the Gaussians).

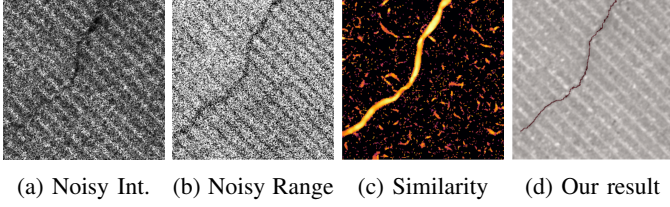


Fig. 4: Noisy FIND example ($s = 0.5$) with speckle noise on intensity and Gaussian noise on range. The graph similarity remains concentrated around the main crack.

TABLE IV: Quantitative comparison on the two annotated Vaches Noires patches.

Image	Method	JACCARD $\uparrow$	TVERSKY $\uparrow$	WASSERSTEIN $\downarrow$
Test1	Ours	0.47	0.52	34 px
	U-Net + T.L.	0.50	0.60	38 px
Test2	Ours	0.29	0.32	81 px
	U-Net + T.L.	0.34	0.36	60 px

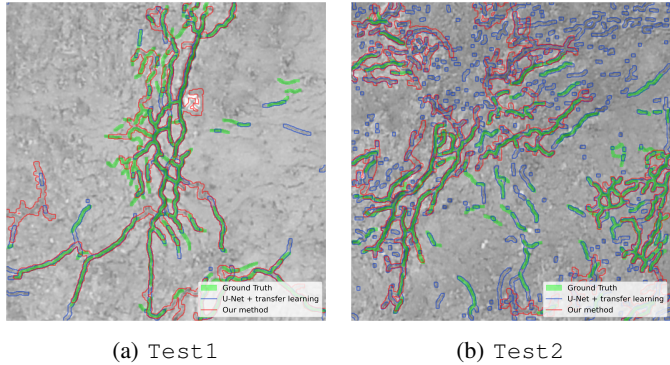


Fig. 5: Vaches Noires patches. Ground truth: Green; ours: Red; U-Net with transfer learning: Blue.

$\Sigma = \{1, 3, 5, 7, 9\}$, $R = 3$ and $\tau = 0.3$ for both evaluation patches.

We compare against a U-Net baseline adapted by transfer learning [25] on an annotated 3760×2058 pixel image from the same area, following the common strategy used when only a small amount of domain-specific annotation is available. This ground truth annotation required approximately two days of work by an expert geophysicist. The proposed method uses only the visible image and no training annotation.

The supervised baseline gives slightly higher overlap scores

on both patches, which is expected because it was adapted on a nearby annotated image. Our method uses no annotation and gives a lower WASSERSTEIN distance on Test1. These two patches are useful transfer tests, but they are not enough to establish performance across domains.

We also apply the method to a fractured limestone wall at the Palais des Papes, using an orthoimage and a photogrammetric depth map. Figure 6 shows that the main fracture is recovered without retraining; minor cracks are missed where depth resolution and contrast are insufficient.

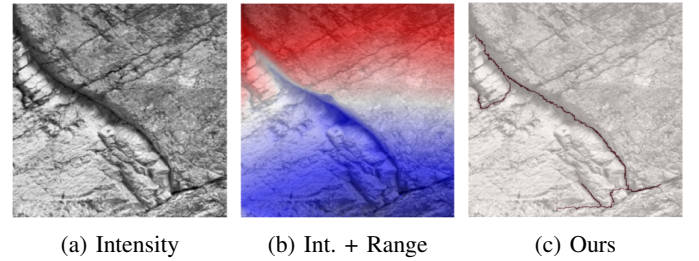


Fig. 6: Palais des Papes wall: Intensity, depth overlay and extracted skeleton.

CODE AND DATA AVAILABILITY

Code, notebooks and non-public data are available in the provided Frangi-EUVIP GitHub repository: <https://github.com/Ayana-Inria/Frangi-EUVIP>. FIND is public [6].

V. CONCLUSION

We presented a geometric, training-free method for multimodal crack skeletonization. On clean FIND data, the supervised model performs better; under added noise, our graph method degrades more slowly. The geological results show that the method can be applied without retraining, but the evaluation is limited to two annotated patches and one qualitative case.

The editable graph supports semi-automatic annotation. Future priorities are direct comparisons with SAM [26] and CrackSAM [27], sensitivity tests on established datasets [28], and higher-order graph variants [29].

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